BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: 92101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Halogens react with metals to form.

(A) Halides
(B) Oxides
(C) Halogen sulphides
(D) Hydrogenated compounds

Q2. Which one of the following is reducing agent.

(A) Cl_2
(B) $KMnO_4$
(C) Mg
(D) $K_2Cr_2O_7$

Q3. The oxidation number of oxygen in H_2O_2 is.

(A) +2
(B) -2
(C) +1
(D) [illegible in source scan]

Q4. Mass of an element is mostly due to its.

(A) Nucleus
(B) Neutrons
(C) Electrons
(D) Protons

Q5. The vapour pressure of a liquid increases with the.

(A) Increase of pressure
(B) Increase of temperature
(C) Increase of intermolecular forces
(D) Increase of polarity of molecule

Q6. Elements in a _____ have similar chemical properties.

(A) Period
(B) Group
(C) Both A and B
(D) Neither A nor B

Q7. The Mathematical formula of density is.

(A) d = m x v
(B) d = v/m
(C) d = m/v
(D) d = v x m

Q8. One mole of glucose ($C_6H_{12}O_6$) has a mass of.

(A) 178g
(B) 179g
(C) 180g
(D) 182g

Q9. The symbol of Boron is.

(A) Br
(B) Bi
(C) B
(D) Be

Q10. The bond in $CaCl_2$ is.

(A) Covalent bond
(B) Ionic bond
(C) Co-ordinate covalent bond
(D) Hydrogen bonding

Q11. Which one of the following can be broken down into simpler substances.

(A) Ammonia
(B) Oxygen
(C) Sulphur
(D) Iron Filling

Q12. The Metallic characters _____ from left to right in periods.

(A) Increases
(B) Decreases
(C) Remains constant
(D) First decrease then increase

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Write the electronic configuration of $_{10}Ne$ and $_{14}Si$ (silicon)?

(ii) Why are the densities of gases lower than that of liquids?

(iii) Define average atomic Mass? Write its formula.

(iv) Why NaOH is strong but NH_4OH is weak electrolyte?

(v) Define Modern periodic law? Distinguish between a period and a group in the periodic table.

(vi) What happens during displacement reaction in halogens?

(vii) Find out the number of proton, electron and neutron in the following elements. (a) $^{23}_{11}Na$ (b) $^{16}_{8}O$

(viii) Give the reason that why dative bond is always polar?

(ix) Differentiate between condensation and evaporation?

(x) Calculate the molarity of 50.0cm³ of solution containing 7.50g of CH_3COH.

(xi) What is the difference between saturated and super saturated solution?

SECTION "C"

Maximum Marks: 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [3] Define chemistry? Write down the contribution of Ibne-Sina in chemistry.

(b) [4] Write down at least four typical properties of solids?

QIV.

(a) [3] What are the factors affecting Electronegativity?

(b) [4] How many moles of H_2SO_4 are present in 0.500dm³ of 0.150M H_2SO_4 solution.

QV.

(a) [3] Draw the structure and shape of sub-shells.

(b) [4] Differentiate between the process of oxidation and reduction. Write an equation to illustrate each.

QVI.

(a) [4] What is the main distinction between ionic and covalent bonding? Explain your answer With suitable examples.

(b) [3] Balance and complete the following equation of Halogens. (i) $H_2 + Cl_2$ (ii) $O_2 + F_2$ (iii) $2P + 3Cl_2$